

“Tracing the Thermal Footprint of Urban-Industrial Expansion: A Satellite-Based Analysis of Surface Urban Heat Island Dynamics in Savar, Bangladesh”

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ABSTRACT

This study examines long-term trends in the Surface Urban Heat Island (SUHI) in the peri-urban region of Savar Upazila, located adjacent to the rapidly growing city of Dhaka, Bangladesh, over the period 2000 to 2025. Landsat imagery was acquired during the warm season and used to develop percentile-based LST metrics (p75, p90). Surface classification was conducted using spectral indices, hotspot persistence was analyzed, and ERA5-Land was incorporated for atmospheric contextualization. Results indicate that SUHI exhibited a non-linear evolution; mean SUHI declined during 2005-2011 before increasing sharply in 2015-2016 and remaining elevated afterward. The scaling relationship between mean and extreme SUHI was weak, suggesting threshold-like amplification. Additionally, the decline in spatial variation indicates increasing thermal homogenization, while residual analysis after accounting for atmospheric effects confirms that most of the heat island intensity was surface-driven. The results further show that persistent hotspots are concentrated along major development corridors within the peri-urban landscape under monsoonal influence.

1. Introduction:

Urban Heat Island (UHI) effects have increased worldwide due to changes made to the Earth's surface as a result of rapid urban development. This process occurs because vegetation is replaced with man-made structures, which increases the amount of stored heat and alters how the surface interacts with its environment in terms of energy exchange. As a result, urban locations generally experience warmer temperatures than those found in rural environments (Zhao et al., 2014). The thermal differences that exist between an urban location and its rural surroundings are further modified by seasonal variations in humidity, cloud cover, and precipitation in monsoon-influenced cities.

One of the world's most densely populated cities, Dhaka, has undergone rapid and largely uncontrolled growth during the last few decades. Research using satellite data has demonstrated increasing levels of surface heat island intensity with the expansion of built-up areas and the decrease in vegetation cover (Uddin et al., 2022; Dewan et al., 2021a). However, it is likely that due to the transitional peri urban area, Savar Upazila has also experienced similar thermal impacts. In addition, ground-based studies have shown that canopy-layer heat island intensity has significantly increased and that there are strong seasonal and daily variations in temperature, with the highest values occurring in winter and at night (Tabassum et al., 2024). It appears that both land surface modifications and atmospheric conditions contribute to the urban thermal environment in Bangladesh.

However, very little research has examined the peri-urban transition zone of Dhaka. On the northwest side of Dhaka lies Savar Upazila, which is undergoing rapid industrial growth as well as mixed land covers such as developed land, agricultural fields, wetlands, and remaining vegetation can be found within this Upazila.

Peri-urban transition zones, therefore, often display fuzzy boundaries between rural and urban areas and a high degree of spatial variation, which can complicate the application of traditional methods for assessing surface urban heat island (SUHI) (Liou et al., 2024; Le et al., 2023). Furthermore, previous research conducted in Bangladesh has primarily focused on average SUHI intensity, with less emphasis placed on extreme thermal behavior and the duration of hotspot persistence.

The ability to differentiate between surface-based enhancement of thermal conditions and atmospheric variability presents a methodological challenge. Prior studies utilizing Landsat-based LST data identified limitations related to atmospheric correction (Ahmed et al., 2013). Additionally, more recent research emphasizes the role of background meteorological conditions in modulating UHI intensity (Tabassum et al., 2024). Therefore, there is a need for a long-term, consistently applied SUHI assessment methodology that incorporates surface characterization, extreme-value metrics, and atmospheric context.

This research explores the evolution of Surface Urban Heat Island (UHI) in Savar Upazila using multi-temporal Landsat imagery over a 25-year time frame. The study utilizes percentile-based Land Surface Temperature (LST) metrics, dominance-based surface classification, hotspot persistence analysis, and ERA5-Land atmospheric context to determine (i) changes in dominant land cover type(s), (ii) the thermal contrast between urban and vegetated areas under monsoonal conditions, and (iii) whether SUHI development is consistent with linear urban scaling theory or exhibits non-linear surface-driven amplification that is independent of atmospheric variation.

2. Literature Review

2.1 Importance and Rationale of UHI Research in Rapidly Urbanizing Regions:

The importance and reason for studying the "Urban Heat Island" (UHI) effect is increasing due to the rapid rate of urbanization around the world and its environmental impacts, in addition to its implications for climate adaptation. The UHI effect causes land surface temperature to rise when natural vegetative surfaces are replaced with impervious man-made materials. This results in higher land surface temperature (LST) in urban areas compared to non-urban or rural areas which eventually emphasize the need for improved quantitative methods for assessing and monitoring SUHI and its driving factors (Liu & Wang, 2023). Consequently, satellite data have been widely used to assess SUHI because LST enables spatially explicit measurement of the urban-rural thermal contrast (Yao et al., 2019). However, the magnitude of SUHI is not only related to the urbanization process but also to changes in vegetation cover and surface moisture in rural areas (Yao et al., 2019).

South Asian countries are experiencing rapid urbanization; therefore, thermal exposure levels are increasing. Dewan et al. (2021) reported a clear relationship between rapid urban development and increasing temperatures in the region, while Uddin et al. (2022) identified significant urban expansion and rising UHI intensity in Dhaka between 2001 and 2017. At the district level, Rahman et al. (2022) demonstrated that in Bangladesh, LULC transitions, expansion of impervious surfaces, and reduced evapotranspiration are significantly associated with UHI dynamics. Nevertheless, Savar Upazila is located on the peri-urban fringe of Dhaka city and represents a rapidly developing industrial and urban landscape. Although Savar Upazila is undergoing substantial land-cover changes, there are limited studies that clearly distinguish surface-driven thermal amplification caused by land-use transformation from atmospheric variability over time.

2.2 Evolution of SUHI: Methods, Drivers, and Interpretative Challenges:

Although SUHI is commonly defined as the difference in LSTs between urban and rural areas, the actual size of the SUHI is greatly influenced by both how the SUHI is measured and which rural areas are used as a reference. Studies by Liou et al. (2024) have shown that different SUHI indicators can have little or no

correlation with each other within the same city. Additionally, Li et al. (2022) demonstrated that using buffer-based, area-based, or land-cover-based definitions for rural reference can result in different SUHI magnitudes, where comparison becomes difficult between studies. Furthermore, Seeberg et al. (2022) pointed out that spatial heterogeneity within a city limits the ability to directly compare thermal signals from urban and rural areas. This is especially true for peri-urban systems such as Savar, which lack clear rural-urban boundaries.

There is growing evidence to support the idea that SUHI development is not linearly related to urbanization. Yao et al. (2019) found that the increase in daytime SUHI associated with global urban expansion was largely influenced by changes in rural vegetation conditions. Therefore, vegetation conditions outside the urban core play an important role in shaping the thermal differences between urban and vegetated surfaces. The relationship between vegetation and UHI, along with the impact of seasonal variations in vegetation and moisture conditions, is also evident in Bangladesh. For example, Rahman et al. (2022) indicated that UHI is influenced by vegetation condition, crop phenology, and transpiration rates. Similarly, Hussain et al. (2023) emphasized that rapid transformation of South Asian cities alters their physical characteristics, including vegetation cover and surface moisture content, which in turn affect their thermal environment. It appears that in monsoon-influenced regions such as Savar, seasonal greening patterns and hydrological cycles could be important factors which may influence thermal contrast between urban surfaces and surrounding vegetation, independent of the rate of structural urban growth.

2.3 Research Problem: Atmospheric Influence and Surface-Driven Amplification:

The main problem when examining SUHI is distinguishing between amplifications caused by surface factors and those resulting from natural atmospheric variability. Several earlier Landsat-based LST studies noted limitations in correcting for atmospheric effects or providing adequate validation (Ahmed et al., 2013; Ogashawara & Bastos, 2012). More recently, researchers have emphasized the importance of understanding atmospheric conditions as a way to better interpret urban thermal dynamics. In particular, Le et al. (2023) identified the relationship between surface properties and atmospheric conditions as critical for understanding urban thermal behavior. Similarly, Dewan et al. (2021b) noted that both surface transformations and climate variability play important roles in shaping warming signals within urban areas.

To address these issues, our study incorporated ERA5-Land data to measure air temperatures at the time of satellite overpass. This approach directly accounts for the potential influence of atmospheric conditions and helps distinguish between surface-controlled amplification and atmosphere-driven variations.

2.4 Limitations and Gaps in Previous Studies:

There are a few important concerns in previous SUHI studies as identified in this reviewed literature. A primary issue has been inconsistencies in the indicators and reference definitions used. SUHI magnitudes have been shown to vary considerably depending on the methodology and the way rural reference areas are defined (Liou et al., 2024; Li et al., 2022). Variations in selected indicators and background definitions can complicate comparisons among studies and may lead to different interpretations of SUHI intensity. A second issue relates to cloud contamination and missing data. Thermal remote sensing is highly sensitive to atmospheric conditions, and cloud contamination frequently results in missing or low-quality LST observations (Li et al., 2022). Also, Liou et al. (2026) demonstrated that missing data can affect the temporal scaling of SUHI estimates, which may influence the ability to reliably assess long-term trends.

Furthermore, many SUHI studies focus primarily on average SUHI intensity rather than extreme upper-tail values and hotspot persistence. Si et al. (2022) emphasizes the importance of quantifying spatial variability and localized heat concentrations, indicating that mean SUHI values alone do not adequately represent the full extent of urban thermal risk. However, Bangladesh-specific research gaps in terms of UHI study remain evident.

Rahman et al. (2022) noted the need for more regionally relevant UHI studies in Bangladesh and recommended incorporating additional climatic variables and landscape metrics beyond those typically derived from simple LST-based methods to better understand urban thermal processes.

3. Data and Methodology

3.1 Study Area:

The study focuses on Savar Upazila located on the north-western peri-urban fringe of Dhaka District, Bangladesh, functioning as a transitional zone between the Dhaka metropolitan core and the adjacent rural floodplain landscapes. The spatial coordinates of the study area lie between 23.74° - 24.04° N latitude and 90.19° - 90.36° E longitude (WGS 84). Savar is characterized by mostly low-lying landscape combined with rivers, wetlands, agricultural fields, industrial estates, and rapidly expanding residential zones. Climatically, Savar experiences a tropical monsoon regime characterized by a hot pre-monsoon season (March-May), monsoon rainfall (June-September), and a relatively cooler dry season (November-February), (Uddin et al., 2022).

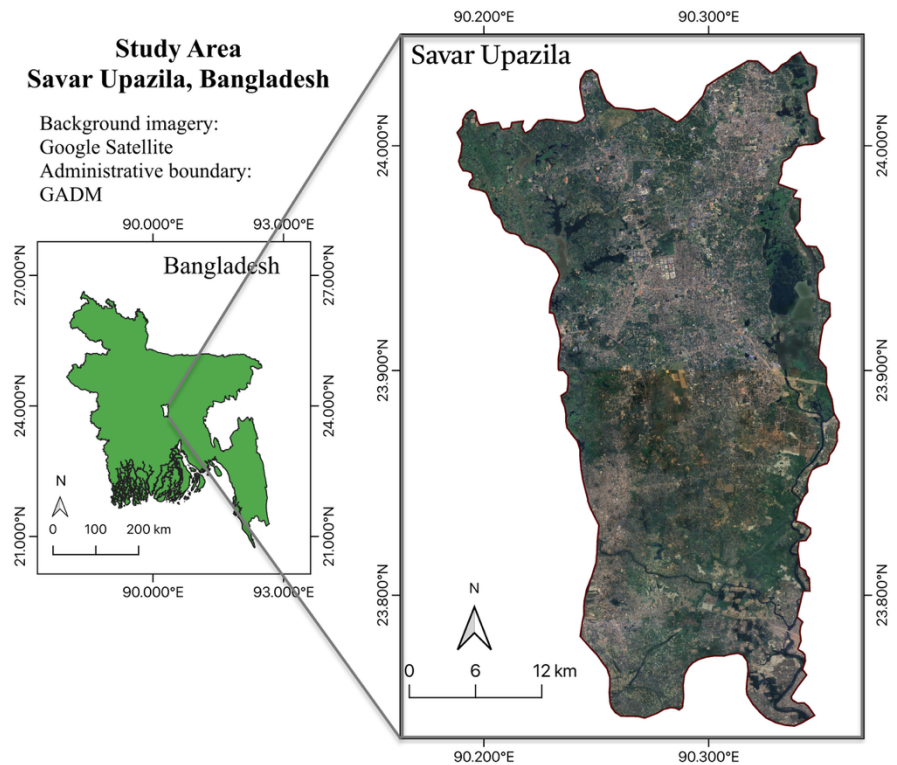


Figure 1. Location of the study area, Savar Upazila, Bangladesh. The inset map shows the national context within Bangladesh, while the main panel displays the administrative boundary of Savar Upazila overlaid on Google Satellite imagery. Administrative boundaries were obtained from GADM.

3.2 Data Sources:

In this study, multi-temporal Landsat imagery was obtained from the United States Geological Survey (USGS) Earth Explorer platform. The analysis covers six temporal blocks spanning 2000 to 2025. Each block represents a two-year composite of warm-season imagery (April-July). A two-year compositing strategy was adopted due to the limitation of cloud-free scenes and to minimize the influence of cloud contamination, data gaps, and scene-level noise.

Table 1. Landsat scenes and quality-controlled image count by temporal block (2000–2025).

Year Block	Time Frame	Satellite Image	Total Scene Taken (quality-controlled pixels, at least 50 to 100%)
2000–2001	April-July	Landsat 7 ETM+	5
2005–2006	April-July	Landsat 5 TM (Landsat 7 ETM+ SLC-off)	4
2010–2011	April-July	Landsat 5 TM (Landsat 7 ETM+ SLC-off)	6
2015–2016	April-July	Landsat 8 OLI/TIRS	12
2020–2021	April-July	Landsat 8 OLI/TIRS	8
2024–2025	April-July	Landsat 9 OLI-2/TIRS-2	27

All Landsat datasets used in this study were atmospherically corrected surface reflectance products (Collection 2 Level-2). The thermal infrared bands were used to derive land surface temperature (LST). In addition, ERA5-Land 2 m air temperature (t2m) data were incorporated to provide background atmospheric context and to examine whether observed SUHI variations were influenced by broader meteorological conditions.

3.3 Pre-Processing and Compositing:

All image processing and analysis were conducted using Google Earth Engine (GEE). Pre-processing steps included cloud masking based on QA bands, seasonal filtering of April-July scenes, and percentile-based compositing. As we found frequent cloud contamination and data gaps during the warm season, a percentile approach was adopted instead of simple mean compositing. Therefore, 75th percentile (p75) was selected to better represent elevated surface temperature conditions while reducing the influence of residual noise and extreme outliers. Compared to mean or median composites, p75 allows our analysis improved tracing of higher-end surface temperature behavior, which is particularly effective for the SUHI analysis.

The p75 percentile was calculated as:

$$p_q(X) = \inf\{x \in \mathbb{R}: F_X(x) \geq q\}$$

where $q = 0.75$ for p75 and $q = 0.90$ for p90, and F_X represents the empirical cumulative distribution function (CDF) of pixel-wise values within each temporal block. Moreover, land surface temperature (LST) and spectral indices were derived from atmospherically corrected surface reflectance products. LST was calculated from Landsat thermal infrared bands using standard brightness temperature (BT) conversion and radiative transfer procedures, incorporating sensor-specific calibration constants.

3.4 Spectral Indices and Land Surface Classification:

Instead using true land use land cover data we classified dominant surface types, using following indices and which were calculated:

$$\begin{aligned} \text{NDVI} &= \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}} \\ \text{NDBI} &= \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}} \\ \text{MNDWI} &= \frac{\text{Green} - \text{SWIR}}{\text{Green} + \text{SWIR}} \end{aligned}$$

Surface dominance was defined using index-based thresholds and remain constant for all year blocks. We used threshold limits for Urban-dominant: $\text{NDBI} > -0.05$ and $\text{NDVI} < 0.25$, Vegetation-dominant: $\text{NDVI} > 0.20$, and Water-dominant: $\text{MNDWI} > 0$ and $\text{NDVI} < 0.25$ and $\text{LST} < 35^\circ\text{C}$. However, water surfaces were excluded from SUHI calculations. To accurately identify persistent water bodies and minimize misclassification, the JRC Global Surface Water dataset was used to delineate long-term water-dominant areas.

3.5 Surface Urban Heat Island (SUHI) Intensity:

In this study, SUHI intensity was calculated as:

$\text{SUHI} = \text{LST}_{\text{urban}} - \text{LST}_{\text{vegetation}}$, Where, urban and vegetation reference pixels were defined based on dominant index classifications. Mean SUHI (p75) was used to represent intensity and extreme SUHI (p90) was used to identify hotspot thresholds, though, hotspots were defined as the upper 10% ($\geq p90$) of SUHI values for each year block.

3.6 Hotspot Persistence and Spatial Density:

Hotspot persistence was assessed by counting the number of periods in which a pixel exceeded the p90 SUHI threshold (range: 0-6). To evaluate spatial consolidation, a 500 m moving window density analysis was conducted to identify clusters of recurrent hotspots over the entire study area. However, persistence classes were categorized as:

- Low (repetition at least 1-2 periods)
- Moderate (repetition at least 3 periods)
- High (repetition at least 4-5 periods)
- Very high (≥ 5 periods)

3.7 Atmospheric Contextualization:

Finally, ERA5-Land 2 m air temperature (t2m) data were extracted for corresponding April-July periods. We used ERA5 data to assess whether observed SUHI trends were attributable to background atmospheric warming or surface-driven amplification processes instead explicitly validating the whole SUHI study.

4. Results

4.1 Index-based land-cover dynamics and urban dominance in Savar:

Figure 2, Table 2 present an overview of the index-based land-cover dynamics in Savar Upazila across six temporal blocks (April-July) from 2000-2001 to 2024-2025. Dominant surface characteristics were derived using consistent spectral indices that represents urban, vegetation, and water signatures. The results indicate a clear long-term reorganization of surface dominance, marked by episodic intensification of urban signatures, substantial variability specially in water-dominant surfaces, and persistent but dynamic vegetation dominance.

Table 2: Area (km²) and percentage (%) of dominant land-cover classes in Savar Upazila (2000–2025)

Year block	Urban (km ²)	Urban (%)	Vegetation (km ²)	Vegetation (%)	Water (km ²)	Water (%)
2000–2001	27.25	7.83	186	53.42	80.48	23.11
2005–2006	97.79	28.08	220.47	63.32	29.05	8.34
2010–2011	96.22	27.63	188.9	54.25	59.72	17.15
2015–2016	48.9	14.04	225.12	64.65	71.93	20.66
2020–2021	99.41	28.55	236.52	67.93	23.09	6.63
2024–2025	83.64	24.02	202.63	58.19	69.65	20

Urban-dominant surfaces exhibit an overall upward tendency through time. In 2000-2001, urban-dominant signatures cover approximately 28 km² (about 8% of the study area), increasing to nearly 85 km² (around 23 to 24%) by 2024-2025). This temporal trajectory is distinctly non-

monotonic, with two phases of intensified urban dominance: an early expansion occurred in between 2000-2001 and 2005-2006, and a second peak around 2020-2021, when urban-dominant surfaces reach their maximum extent (≈ 100 km²). A temporary contraction is noticed in 2015-2016 which interrupts this trend, indicating short-term moderation in urban surface expression rather than a reversal of long-term change.

On the other hand, vegetation-dominant surfaces remain the most extensive class throughout the study period but present a pronounced inter-block variability. Periods of strong urban surface expression (2005-2006 and 2020-2021) coincide with elevated vegetation-dominant extent (≈ 220 and 237 km²), that indicates the increases in urban-dominant signatures do not uniformly correspond to reductions in vegetation-dominant surface signals. Instead, these periods are primarily characterized by contractions in water-dominant surfaces (≈ 29 km² in 2005-2006 and ≈ 23 km² in 2020-2021), suggesting a redistribution among dominant surface signatures rather than direct urban-vegetation displacement. Following the 2020-2021 peak, vegetation dominance weakens in 2024-2025 (≈ 203 km²), which eventually highlights the non-linear nature of index-based surface characterization.

Water-dominant surfaces show pronounced temporal variability and a consistent inverse association with urban surface dominance. Periods of intensified urban dominance correspond to the clear reductions in water-dominant extent, whereas intervals of reduced or stabilized urban dominance (2010-2011 and 2015-2016) coincide with recovery of water-dominant surfaces, exceeding 70 km² in some blocks. This pattern indicates that temporal reconfiguration of surface dominance in Savar is facilitated primarily through shifts in water-related spectral signatures rather than uniform displacement of vegetation-dominant surfaces.

4.2 Temporal Changes in LST and Urban-Vegetation Contrast (April-July, p75):

The p75 LST maps reveal obvious inter-block changes in the dominant temperature classes. Blocks 2000-2001, 2005-2006, and 2015-2016 (**Figures 3a, 3b, 3d**) are characterized by widespread dominance of the 36-44 °C class, whereas noticeable localized areas reaching higher temperature ranges. Most of the areas are fall in under the class 36-44 °C and their common hot zone in central Savar (e.g., Savar city area, Bipail), Ashulia industrial zones and Amin Bazar nearby Dhaka at southeastern part of the Savar upazila.

However, block 2000-2001 shows an extensive heat area at the western and northwestern part of the study area. In contrast, 2010-2011 (**Figure 3c**) shows a clear shift toward cooler classes (28-36 °C and 36-44 °C), where high-temperature categories have become spatially limited. The later blocks (2020-2021 and 2024-2025; **Figures 3e, f**) exhibit stable but more spatially fragmented warm signatures, which indicate the

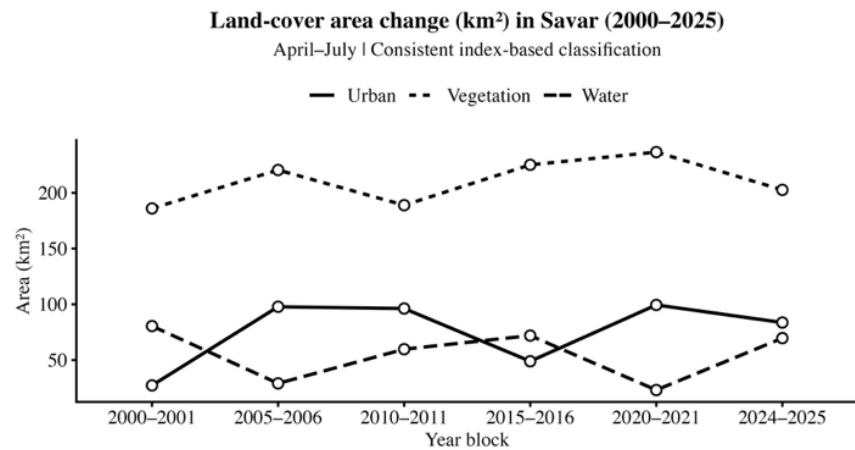


Figure 2. Land-cover area change (km²) in Savar Upazila from 2000 to 2025 based on April–July composite imagery using consistent index-based classification. Trends are shown for urban, vegetation, and water dominance classes across six temporal blocks.

persistence of elevated p75 LST in localized areas rather than across the broader landscape, which eventually does not satisfy expectation of urban growth insight. However, the distinct urban, climatic, and geographic conditions establish Savar Upazila as a climatically sensitive peri-urban system, where land-cover reconfiguration including spectral variations can rapidly amplify surface thermal extremes.

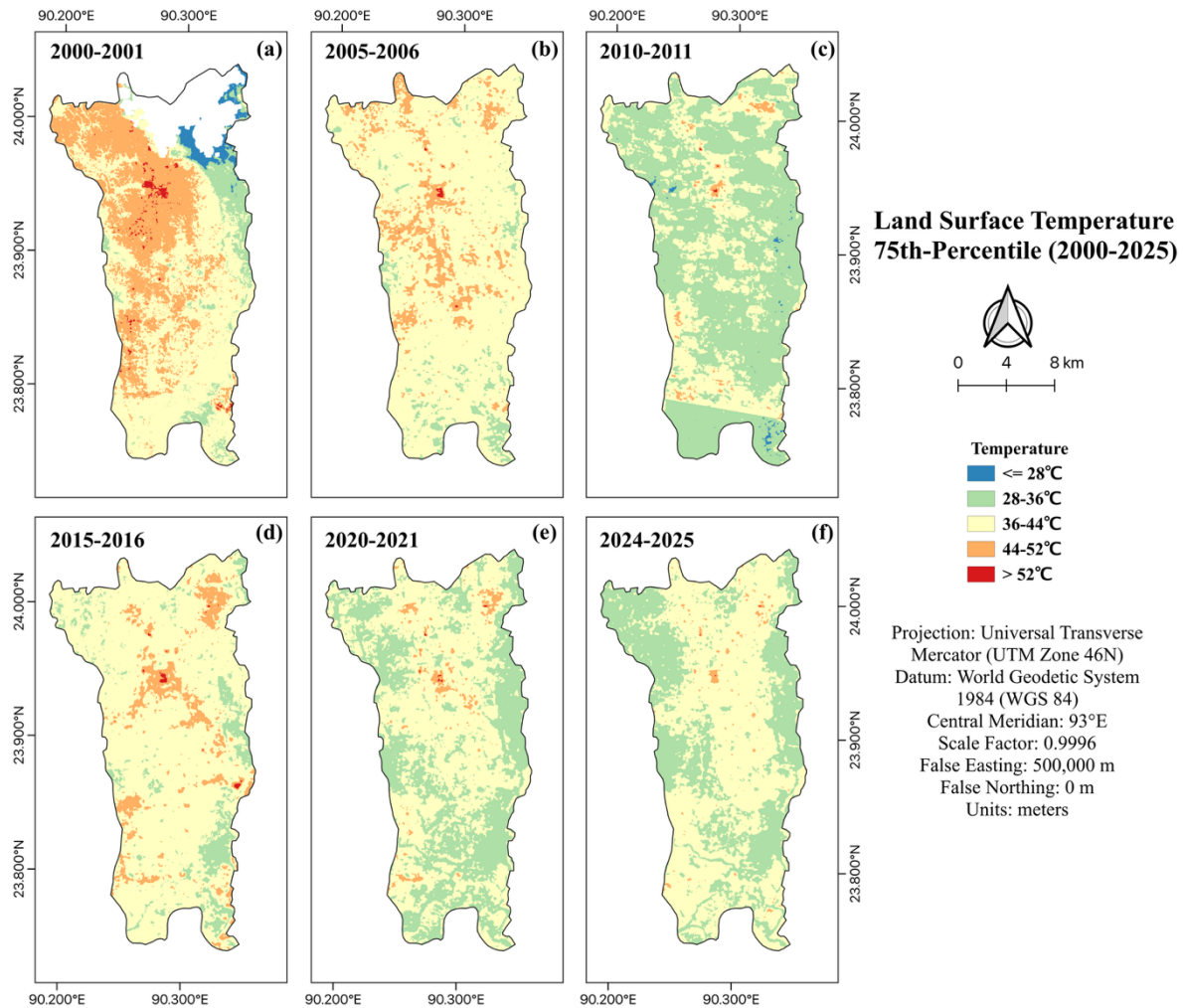


Figure 3. Spatiotemporal evolution of 75th-percentile land surface temperature (LST) in Savar Upazila during 2000-2025 based on April-July Landsat composites. Panels (a-f) represent six two-year temporal blocks. Temperature classes (°C) highlight elevated thermal conditions associated with surface characteristics. Projection: UTM Zone 46N, WGS 84.

Both urban and vegetation-dominant surfaces demonstrate their highest mean p75 LST values in 2000-2001, followed by a pronounced decline toward 2010-2011, again a secondary increase in 2015-2016, and comparatively lower but also stable values in 2020-2021 and 2024-2025 (Figure 4; Table 3). This temporal evolution is broadly similar for both surface classes i.e., urban and rural like, despite persistent differences in absolute temperature levels, indicating that

Table 3. Mean 75th-percentile land surface temperature (LST_p75, °C) for urban and vegetation-dominated surfaces and corresponding SUHI intensity

Year block	UrbanMean	RuralVegMean	Delta
	LST_p75_C	LST_p75_C	LST_C
2000-2001	45.67	42.73	2.94
2005-2006	42.28	40.30	1.98
2010-2011	36.68	34.76	1.91
2015-2016	43.90	39.16	4.73
2020-2021	39.62	36.36	3.27
2024-2025	40.31	36.44	3.87

periods of elevated or suppressed p75 LST affect urban and vegetation-dominant signatures concurrently. The coexistence of high p75 LST in both classes during the early period, followed by simultaneous cooling toward 2010-2011, produces a non-monotonic thermal trajectory that does not directly mirror the progressive urban expansion which was documented in the first section. In contrast to this parallel movement in absolute p75 LST, the urban-vegetation thermal contrast (Δ LST) displays a distinct temporal pattern. Δ LST remains relatively small in the earlier and intermediate blocks (2005-2006 and 2010-2011, $\approx 2^\circ\text{C}$) but increases sharply in 2015-2016 ($\approx 4.7^\circ\text{C}$) and remains elevated in the most recent periods (2020-2021 and 2024-2025, $> 3^\circ\text{C}$), even as absolute p75 LST levels stabilize or decline.

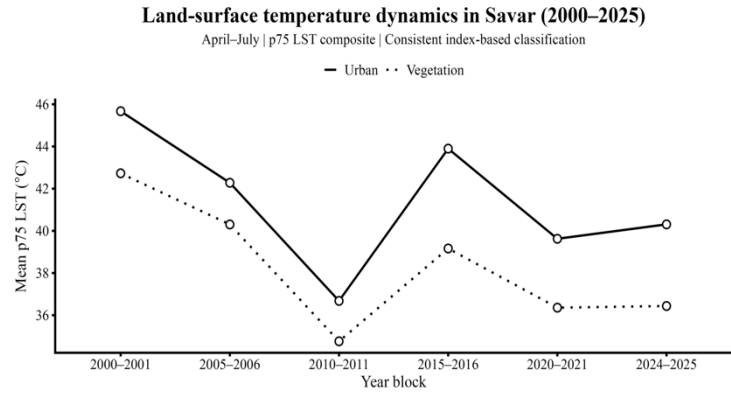


Figure 4. Temporal dynamics of mean 75th-percentile land surface temperature (p75 LST) for urban and vegetation-dominated surfaces in Savar Upazila (2000–2025) derived from April–July Landsat composites using consistent index-based classification.

4.3 SUHI Intensity, Extremes, and Hotspot Persistence (April–July, p75):

Our findings of SUHI behavior in Savar shows episodic amplification rather than long term steady intensification with a clear separation between changes in the central tendency and the upper tail. Mean SUHI derived from p75 LST declines from 2.94°C (2000–2001) to ~ 1.9 to 2.0°C (2005–2011), before rising sharply to a maximum of 4.73°C

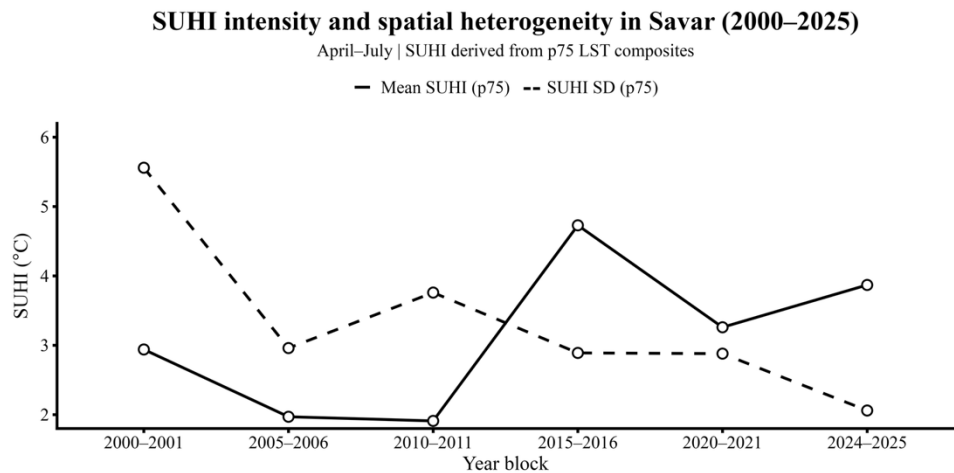


Figure 5. Temporal evolution of mean SUHI intensity (p75) and spatial heterogeneity (standard deviation of p75 SUHI) in Savar Upazila (2000–2025), derived from April–July Landsat composites.

(2015–2016) and remaining elevated through the recent blocks, i.e., 2020–2021 (3.26°C) and 2024–2025 (3.87°C) (**Fig. 5; Table 4**). In parallel, spatial heterogeneity behaves in the opposite direction; **Fig. 5 and Table 4** show SUHI SD is highest in 2000–2001 (5.56°C) and progressively compresses to $\sim 2.9^\circ\text{C}$ (2015–2021) and 2.06°C (2024–2025). This reveals a clear contrast in SUHI behavior where recent periods show higher average SUHI while spatial differences become smaller.

The block-wise SUHI maps reveal a systematic reorganization of the spatial structure of urban heat across Savar (**Fig. 6**). In the earlier blocks, high SUHI values are largely restricted to a limited number of discrete locations, e.g., dense industrial zones, high impervious areas, embedded within extensive areas of weak to moderate SUHI, and these early blocks reflect a highly heterogeneous heat pattern. In contrast, later blocks are characterized by increasingly coherent and spatially contiguous high-intensity SUHI zones, most

prominently along the central development corridor and within the north-eastern sector, that indicates a transition from localized to more widespread urban heat expression. However, from this progression an important deviation is also evident in 2010–2011, where extensive weak or negative SUHI conditions coexist with a small number of pronounced hotspots.

Table 4. Summary of mean rural and urban LST (p75, °C), mean SUHI (p75), extreme SUHI (p90), and SUHI spatial heterogeneity (standard deviation of p75 SUHI) across six temporal blocks in Savar Upazila from 2000 to 2025.

Year block	RuralMean_LST p75_C	UrbanMean_LST p75_C	Mean_SUHI p75_C	SUHI p75_P90_C	SUHI p75_StdDev_C
2000/2001	42.72	45.67	2.94	8.7	5.56
2005/2006	40.3	42.27	1.97	5.68	2.96
2010/2011	34.76	36.67	1.91	7.19	3.76
2015/2016	39.16	43.89	4.73	8.18	2.89
2020/2021	36.35	39.62	3.26	7.18	2.88
2024/2025	36.43	40.3	3.87	6.3	2.06

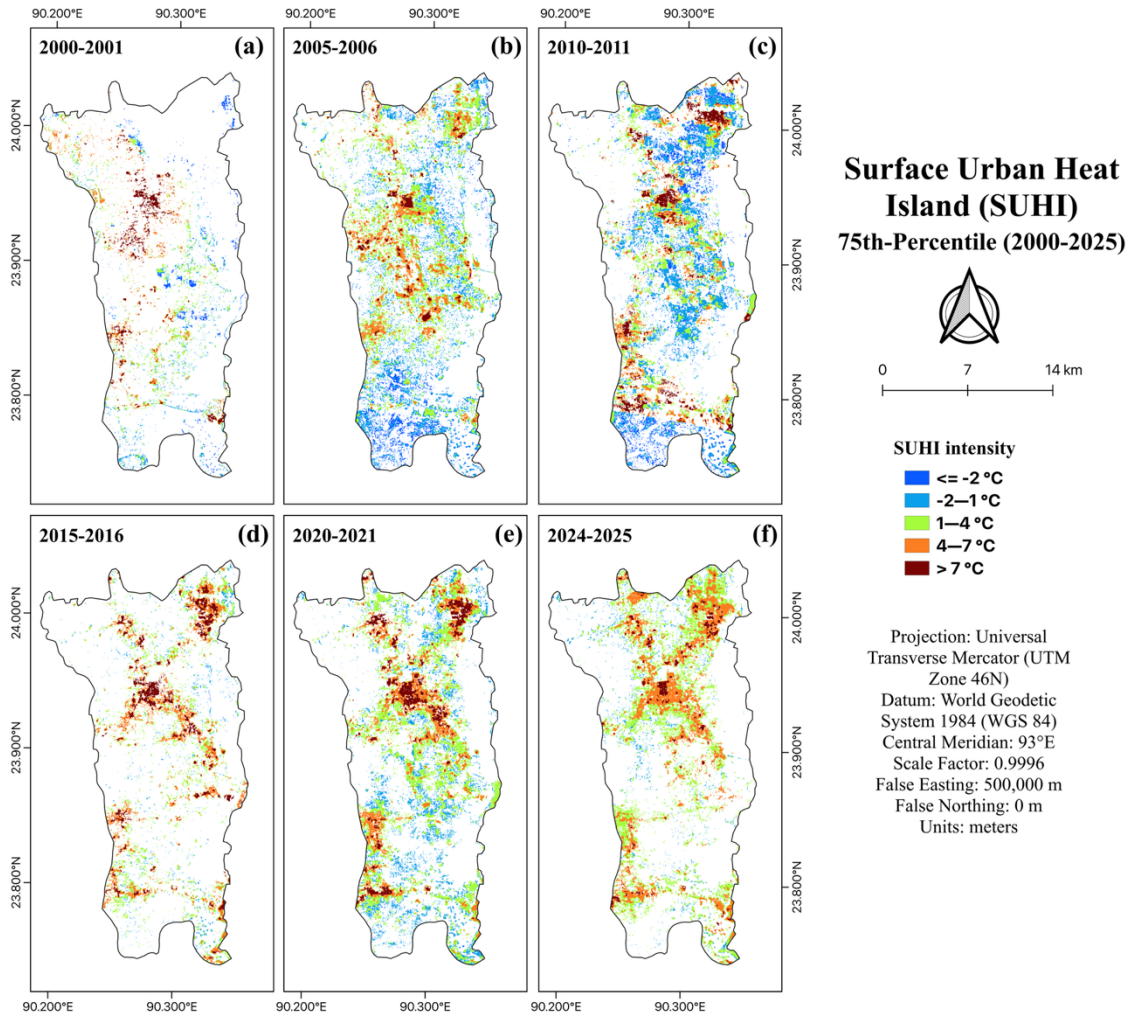


Figure 6. Spatiotemporal evolution of 75th-percentile SUHI intensity in Savar Upazila during 2000–2025. Panels (a–f) represent six two-year time blocks derived from April–July p75 LST composites using consistent dominance-based classification. Warmer colors indicate stronger urban thermal amplification.

The extreme SUHI values show that average urban heat and the strongest heat hotspots do not always change together. In some years, the overall SUHI is relatively low, but very hot local areas still exist. For example, in 2010–2011, the average SUHI is one of the lowest in the study period (1.91 °C), yet the extreme SUHI value (p90 = 7.19 °C) remains high (Fig. 7). This means that even when the city is not very hot overall, a few places can still experience very intense heat. This pattern shows that extreme SUHI does not increase in a smooth or proportional way as average SUHI changes. This nonlinearity is further supported by the weak scaling between mean SUHI and p90 SUHI (Fig. 7). The regression relationship ($P90 = 5.96 + 0.40 \times \text{Mean}$) explains little variance ($R^2 = 0.15$; $p = 0.447$), which denotes that extreme SUHI does not increase in a simple linear proportion to mean SUHI.

Scaling Between Mean and Extreme SUHI (P90) Intensity Across Time

Regression-based assessment of linear versus nonlinear SUHI amplification

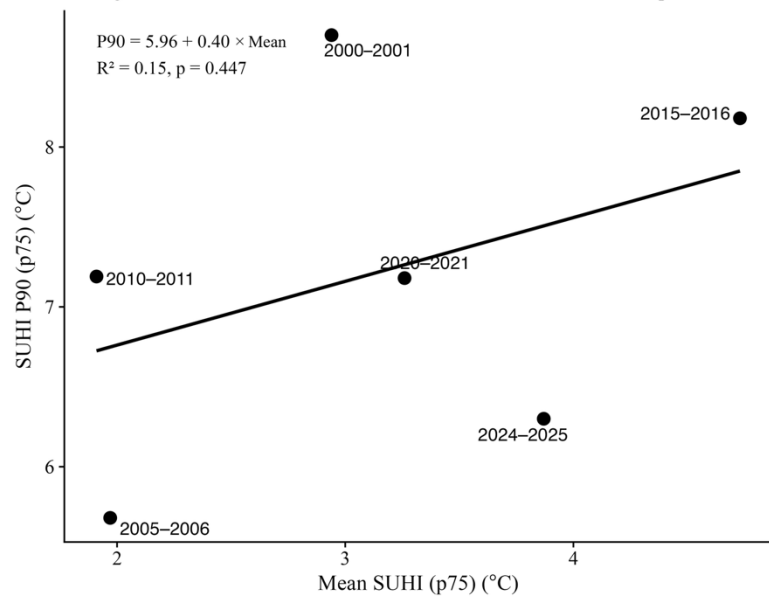


Figure 7. Scaling relationship between mean SUHI intensity (p75) and extreme SUHI intensity (p90) across temporal blocks (2000–2025) in Savar Upazila. The regression line illustrates the weak linear association between mean and extreme SUHI, indicating potential nonlinear amplification.

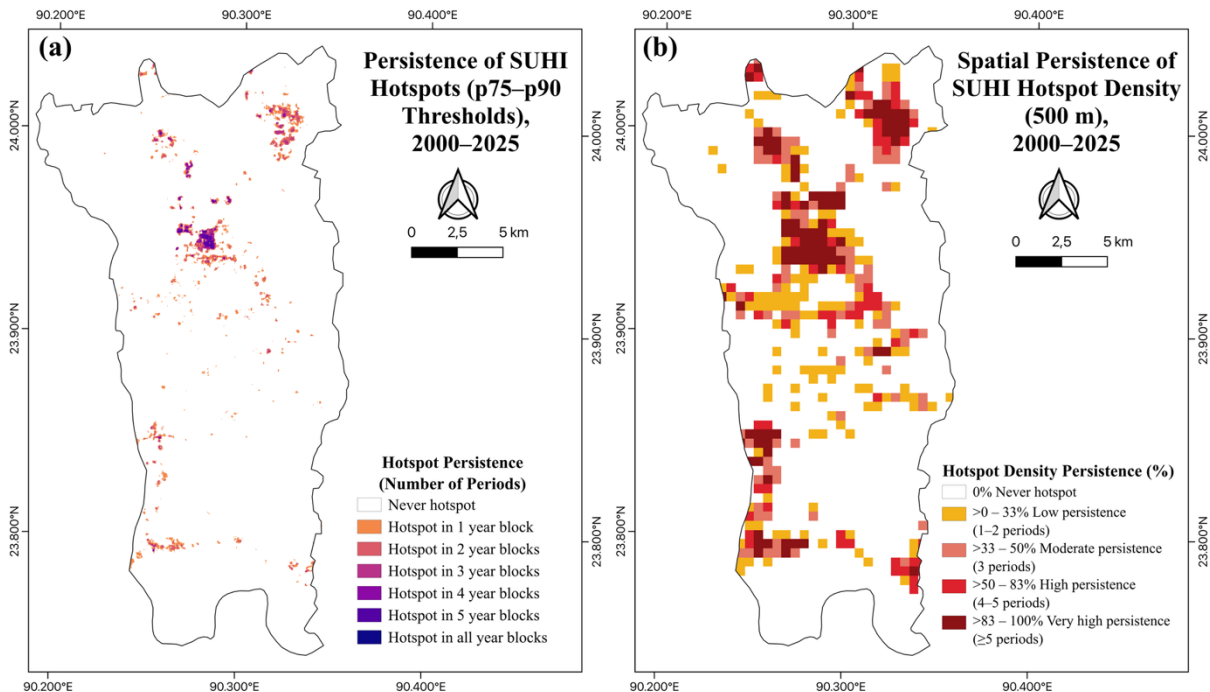


Figure 8. Persistence analysis of SUHI hotspots in Savar Upazila from 2000 to 2025. Panel (a) shows the number of temporal blocks in which each pixel exceeded the p75–p90 SUHI threshold. Panel (b) presents the spatial density of hotspot persistence aggregated within a 500 m grid, highlighting zones of recurrent thermal amplification.

We also have conducted hotspot persistence analysis, and which indicates that these extremes are spatially anchored rather than randomly distributed (Fig. 10). The pixel-level persistence map shows that most hotspot pixels occur in only one or two blocks, but a smaller set of multi-block cores recur repeatedly, especially within the central zone, north-eastern cluster and southeastern cluster (Fig. 8a). The 500 m hotspot-density persistence surface further strengthens this inference through identifying a central north-south belt with moderate-to-very high persistence and additional persistent nodes toward the margins (Fig. 8b).

5. Discussion

5.1 Urban Expansion and Surface-Dominance Reconfiguration:

Land-cover dynamics in Savar between 2000 and 2025 exhibit a non-linear, mosaic-like reconfiguration of urban, vegetation, and water-dominant surfaces that departs from classical models of monotonic urban growth. Although urban-dominant surfaces increased substantially, this expansion did not consistently coincide with vegetation loss. Vegetation dominance remained high or increased during major urban growth phases (e.g., 2005-2006 and 2020-2021), contradicting a simple urban-vegetation substitution model.

This apparent contrast reflects both the peri-urban character of Savar and the dominance-based classification framework. In the Dhaka fringe, built-up expansion commonly coexists with bare land, cropland, homestead vegetation, and tree cover, producing a mixed-use mosaic in which vegetation can remain spectrally dominant despite ongoing urbanization (Ullah et al., 2019; M. Rahman et al., 2022). The April-July observation window further coincides with early monsoon greening, i.e., as a tropical monsoon-controlled climate region, the area experiences abundant growth of lush green vegetation during the early monsoon season (Awasthi et al., 2023), which enhances NDVI dominance even within developing urban areas. Consequently, increases in vegetation dominance indicate surface greenness amplification rather than the absence of urban encroachment. However, the findings of inverse fluctuations between urban and water dominance are primarily driven by seasonal hydrological variability and the dominance-based spectral framework, rather than repeated physical infilling or restoration. Though the water dominant surface trajectory shows comparatively contrasting with strong fluctuations. This pattern may include wetland infilling in Savar, Gazipur, and eastern Dhaka, where low-lying lands are favorably converted for residential & industrial development (Ullah et al., 2024; Sultana, 2009).

Inter-annual differences in surface moisture during the pre-monsoon to early monsoon period alter NDVI, NDWI, and NDBI contrasts, causing transitional pixels to shift dominance. The temporary decline in urban dominance in 2015-2016 therefore reflects climate-sensitive suppression of urban spectral signals under wetter conditions, with concurrent increases in vegetation and water dominance which are linked to monsoon-driven regrowth, surface saturation, seasonal inundation, and transitional construction stages. Similar non-monotonic behavior has been reported in monsoon-influenced peri-urban systems where surface conditions fluctuate despite stable land use. Periods of simultaneously high urban and vegetation dominance indicate strong spatial heterogeneity across Savar, where core industrial and residential intensification coexists with persistent cropland and managed homestead vegetation (Hassan & Southworth, 2017). Because this study captures prevailing spectral-thermal surface properties relevant to LST and SUHI rather than fixed land-use classes, or transitional pixels may shift seasonally without implying structural land-use change (Zhou et al., 2014).

Independent ERA5-Land analysis shows no systematic anomaly in background air temperature during this period, indicating that these surface-dominance shifts are not driven by external atmospheric forcing. Together, index-based landcover dynamics' results show that urban expansion in Savar redistributes dominance among urban, vegetation, and water surfaces preferentially transforming water bodies and modulating vegetation signals without producing a uniform decline in green surfaces. This surface reconfiguration provides essential context for interpreting thermal behavior and SUHI amplification which are examined in the LST section.

5.2 Nonlinear Surface Thermal Dynamics:

The evolution of land surface temperature (LST) and surface urban heat island (SUHI) intensity in Savar between 2000 and 2025 have revealed a distinctly nonlinear thermal response to urban expansion, and these strange patterns strongly modulated by monsoon-driven climate variability and peri-urban surface processes. Contrasts found in return to the linear urban warming expectations, the highest p75 LST values for both urban and vegetation-dominant surfaces occur in the early period (2000-2001). This pattern is unlikely to reflect urbanization effects and instead points to surface energy balance conditions that is associated with the pre-monsoon heat stress and reduced surface moisture availability, operating under broadly comparable atmospheric temperatures. However, ERA5-Land reanalysis confirms the absence of monotonic background air-temperature warming, which provides the insights that early extreme LST reflects surface-state sensitivity rather than long-term climatic forcing. Moreover, similar masking or reversal of urban thermal signals due to interannual climate variability has been reported in other tropical monsoon settings (Li & Bou-Zeid, 2013).

We also found the parallel cooling of urban and vegetation surfaces toward 2010-2011 further underscores the dominance of shared external forcing, likely linked to enhanced cloud cover, rainfall, or soil-moisture recharge during the warm season, i.e., analysis periods. Essentially, this synchronized LST behavior does not refute urban thermal effects but instead highlights that absolute LST trajectories alone are insufficient to isolate urban influence. Besides, the urban-vegetation contrast (Δ LST) provides a more robust indicator of surface differentiation under common atmospheric conditions. From 2015 onward, Δ LST increases sharply, despite stabilization or partial decline in absolute LST. This divergence is not accompanied by rising ERA5 air temperatures, whereas it indicates a transition toward surface-controlled thermal differentiation (**fig. 4**). Processes such as impervious surface consolidation, reduced evapotranspiration, and declining surface moisture increasingly dominate the urban thermal response, eventually, producing a step-change in SUHI intensity from $\sim 2^\circ\text{C}$ to $>4.5^\circ\text{C}$. Nevertheless, significant shift after 2015 has altered land-atmosphere coupling rather than regional climate warming and this findings are consistent with reported decoupling between SUHI magnitude and background climate in Dhaka and other rapidly urbanizing South Asian cities (Dewan, Kiselev, & Botje, 2021b ; Uddin et al., 2022).

5.3 SUHI Amplification in a Monsoon-Driven Urbanizing Landscape:

SUHI amplification in Savar is further characterized by episodic and threshold-like behavior rather than gradual scaling with urban extent. We also considered tail behavior whether the mean SUHI (p75) can be explained by p90 SUHI values. However, finding weak relationship between mean SUHI and extreme SUHI (p90; $R^2 \approx 0.15$); **fig. 7** indicates that average conditions poorly predict tail behavior. Our findings about early periods exhibit high spatial heterogeneity ($SD \approx 5.6^\circ\text{C}$), reflecting a highly fragmented landscape with sharp contrasts between cool wet surfaces and localized hot built-up pockets. In contrast, later periods show declining variability ($SD \approx 2.1^\circ\text{C}$ by 2024-2025) alongside elevated mean SUHI.

However, these findings can be attributed to the land surface dynamics in Savar from 2000 to 2025. In early periods, Savar was considerably characterized by a fragmented peri-urban landscape composed of dispersed settlements, exposed bare surfaces, agricultural fields, wetlands, and relatively limited industrial development. This heterogeneous surface configuration created strong local contrasts between cooler vegetated or water-dominated areas and warmer built-up or bare surfaces, resulting in elevated spatial variability of SUHI during the early period. In contrast, recent years show a more consolidated and spatially continuous urban structure, where built-up and industrial areas have expanded and become more interconnected. This increasing surface homogenization has reduced sharp thermal contrasts across short distances, contributing to lower spatial

variability of SUHI despite higher mean SUHI intensity which are consistent with convergence patterns observed in other tropical and subtropical cities (Manoli et al., 2019; Zhou et al., 2014).

Despite declining the overall variability, hotspot persistence analysis reveals stable thermal cores in central Savar (e.g., Savar City, Ashulia, Bipail, Amin Bazar) across multiple periods. The recurrence of these hotspots independent of interannual climatic fluctuations indicates structural controls linked to fixed land uses and urban form, such as industrial facilities, dense impervious surfaces, and high heat-storage materials. This explains the apparent contrast observed in 2010-2011, where low mean SUHI coincides with pronounced extremes, likewise, localized thermal risks persist even when citywide averages are muted. Similar separation between mean conditions and hotspot behavior has been documented where SUHI extremes are governed by land-use type and surface configuration rather than urban extent alone (Zhao et al., 2014).

5.4 Background atmospheric conditions at satellite overpass:

Scene-matched ERA5-Land 2-m air temperature at Landsat overpass time shows limited inter-block variability across the six analysis periods (**Fig. 9**: ERA5 overpass-time boxplots and mean $\pm$ SD). Median values and interquartile ranges overlap substantially, indicating broadly comparable atmospheric conditions during the April-July observation window. Mean overpass-time air temperatures remain within a narrow low-30 °C range, with no monotonic warming trend over time, though inter-block differences are present but moderate. The 2015-2016 period exhibits the lowest mean air temperature and the widest spread toward cooler values, while 2010-2011 and 2024-2025 show higher upper-tail values and larger dispersion. However, we used ERA5 analysis to understand whether the blocks represent a uniquely extreme atmospheric background relative to the others.

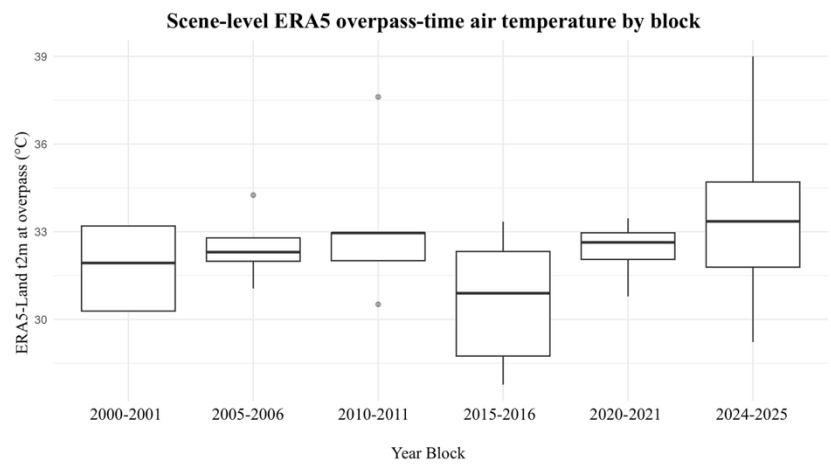
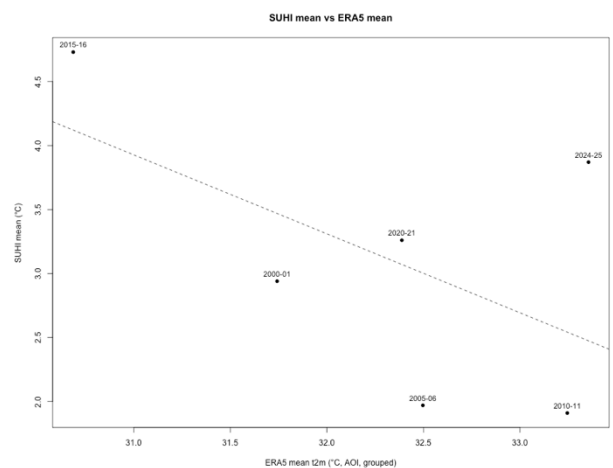


Figure 9. Scene-level ERA5-Land overpass-time air temperature (2 m, °C) across six temporal blocks (2000–2025) in Savar Upazila. Boxplots represent the distribution of atmospheric background conditions corresponding to Landsat acquisition

5.5 SUHI magnitude versus background air temperature:

Comparison of mean SUHI intensity with corresponding ERA5 overpass-time air temperature reveals no systematic linear association (**Fig. 10**: SUHI versus ERA5 overpass-time temperature scatter). Our analysis year blocks experiencing similar atmospheric temperatures display evidently different SUHI magnitudes, and periods with relatively high SUHI do not consistently coincide with the warmest background conditions. Notably, some of the highest mean SUHI values occur under moderate atmospheric temperatures, e.g., 2000-2001 and 2015-2016



blocks, while opposite findings are also noticeable in 2005-2006 and 2010-2011 blocks. These patterns indicate that inter-block SUHI variability is not proportional to contemporaneous background air temperature.

5.6 ERA5-controlled SUHI residuals:

After accounting for background atmospheric temperature using ERA5 data, a clear difference emerges between earlier and later periods (Fig. 11). During 2000-2001, 2005-2006, and 2010-2011, SUHI residuals are close to zero or slightly negative, indicating that observed SUHI during these periods can largely be explained by prevailing atmospheric conditions at the time of satellite overpass. By contrast we found an opposite observation from 2015-2016 onward, where residuals become positive and increase over time, with the largest residual observed in 2024-2025. These significant findings indicate that SUHI in the later periods exceeds what would be expected from background air temperature alone, pointing to an increasing contribution from urban surface characteristics rather than atmospheric forcing.

Figure 10. Relationship between mean SUHI intensity (p75) and ERA5-Land mean 2 m air temperature across temporal blocks (2000–2025) in Savar Upazila. The regression line illustrates the association between atmospheric background temperature and surface-derived SUHI intensity.

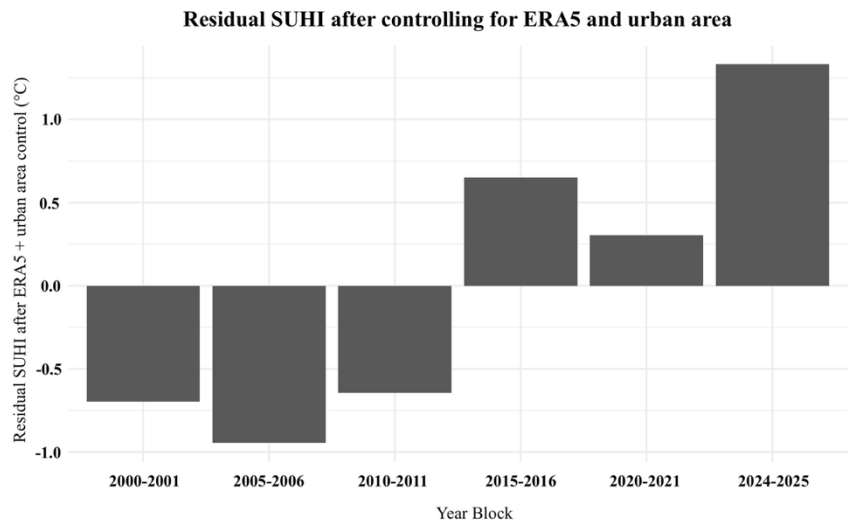


Figure 11. Residual SUHI intensity (°C) after controlling for ERA5-Land air temperature and urban area extent across temporal blocks (2000–2025) in Savar Upazila. Positive residuals indicate surface-driven amplification beyond atmospheric and urban area effects.

6. Conclusion:

In this study, we examined the long-term development of Surface Urban Heat Island (SUHI) dynamics in Savar Upazila, an extremely fast-growing peri-urban industrial area on the outskirts of Dhaka, Bangladesh, during the years 2000-2025. We also analyzed multi-temporal Landsat data using percentile-based land surface temperature metrics, dominance-based surface classification, hotspot persistence analysis, and ERA5-Land atmospheric contextualization to assess surface thermal changes under monsoonal climate conditions.

Our results show that built-up expansion has substantially altered the dominant land cover in the area over time and increased the thermal contrast between urban areas and vegetation. The findings also indicate that SUHI evolution did not increase linearly with urban growth, but rather exhibited episodes of amplified thermal activity and persistent hotspots along major development corridors. Additionally, the declining spatial variation suggests increasing thermal homogeneity primarily due to the rising proportion of impervious surface cover. Finally, atmospheric contextualization further indicates that recent SUHI activity is primarily controlled by surface conditions rather than solely by natural variations in background air temperature.

Our research provides new insights into nonlinear thermal amplification in peri-urban, monsoon-controlled environments and highlights the importance of incorporating extreme-value metrics and hotspot persistence into SUHI assessment frameworks. We recommend that future research include higher-resolution indicators of urban morphology and in-situ meteorological observations to improve understanding of land-

atmosphere interactions and to assess the applicability of these peri-urban thermal patterns across the broader Dhaka metropolitan region.

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